

5. Perform basic Image Handling and processing operations on the image

- **Read an image in Python and Erode an Image using the erode function.**

Aim:

To perform basic image handling operations by reading an image in Python and performing an erosion function using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2

import matplotlib.pyplot as plt

import numpy as np

image =

cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

kernel = np.ones((5,5), np.uint8)
```

```
eroded = cv2.erode(rgb, kernel, iterations=1)
```

```
plt.figure(figsize=(10,5))
```

```
plt.subplot(1,2,1)
```

```
plt.imshow(rgb)
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.subplot(1,2,2)
```

```
plt.imshow(eroded)
```

```
plt.title("Eroded Image")
```

```
plt.axis("off")
```

```
plt.show()
```

Sample Input



Sample Output



Result:

The input image was successfully read using Python and processed using the erosion function of the OpenCV library